

Android Workshop

WebViews

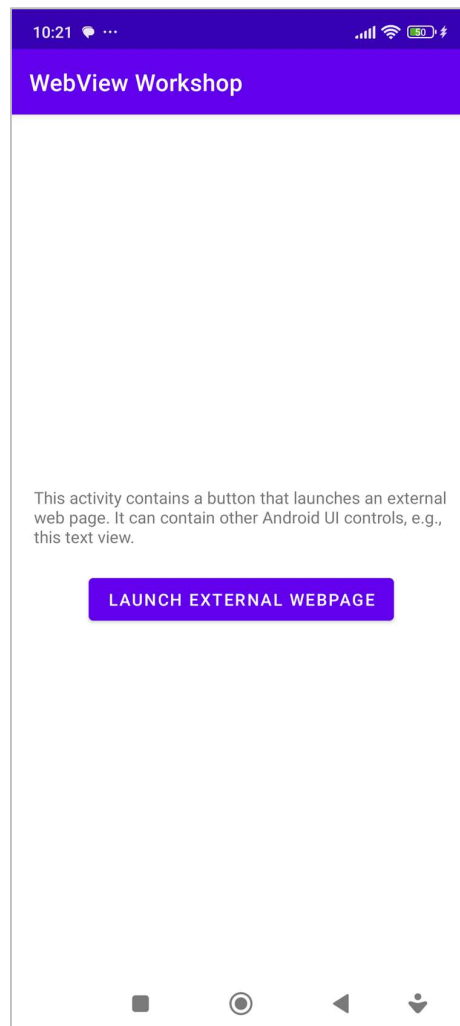
Guide:

- *You are strongly encouraged to type the sample code, instead of copying-and-pasting. In your first round of doing the workshop, you can just copy-and-paste the code to make it work. In the second round, you type the code to understand deeply.*
- *You are also strongly encouraged to practice iterative and incremental development.*

Requirements

In this workshop, we will create an Android app, with the following features.

1. When launched, the app shows the following **Main Activity**.

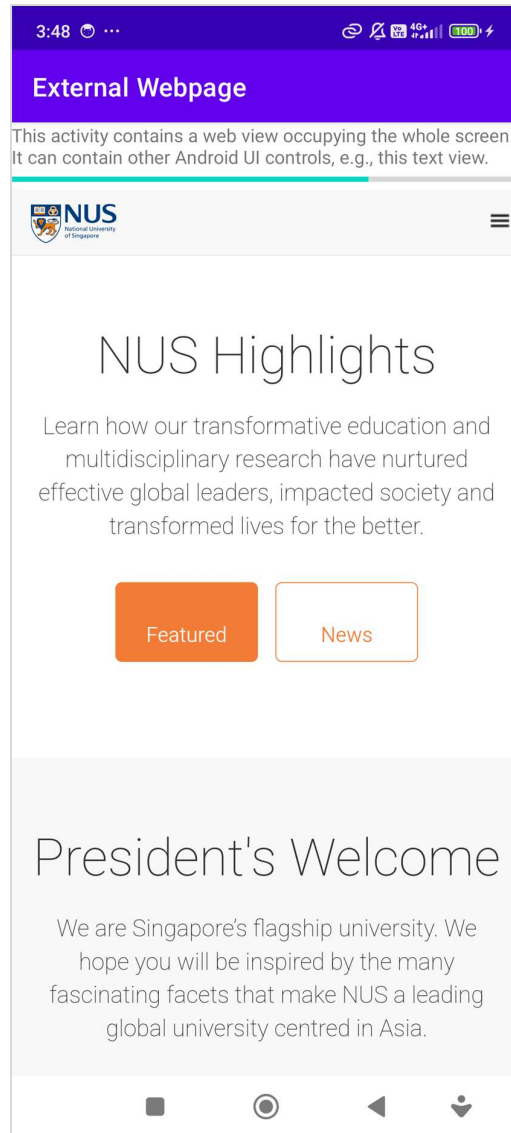


2. When users click “Launch External Website” button, it launches the **WebView Activity**.



This activity contains a whole external web page, e.g., the NUS home page. The URL is <https://nus.edu.sg>. You can choose another URL.

3. When **WebView Activity** is loading the external web page, a Progress Bar appears and shows the loading progress. After the loading completes, the Progress Bar is hidden. As an example, please pay attention to the Progress Bar at the top of the web page in the diagram below.



Set Up Project

1. In Android Studio, create a new project with the following settings.
 - Template: **Phone and Tablet -> No Activity**
 - Name: **External Webpage**
 - Package: **iss.nus.edu.sg.webviews.workshop**
 - Language: **Kotlin**
 - Minimum SDK: **API 24**
 - Build configuration language: **Kotlin DSL**
2. Add the following line to *app/build.gradle.kts* (Module :app), to enable View Binding.

```
...
android {
    ...
    buildFeatures {
        viewBinding = true
    }
}
...
```

Click **Sync Now** to sync Project with Gradle Files

3. Create an **Empty Views Activity**
 - Activity Name: **MainActivity**
 - Generate a Layout File: **check**
 - Layout Name: **activity_main**
 - Launcher Activity: **check**
 - Package name: **iss.nus.edu.sg.webviews.workshop**
 - Source Language: **Kotlin**

- a. Change layout **activity_main.xml** to the following.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical"
    android:gravity="center" >
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="This activity contains a button that launches an
external web page. It can contain other Android UI controls, e.g., this
text view."
        android:layout_gravity="center"
        android:layout_margin="20dp" />

    <Button
        android:id="@+id/launch_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="Launch External Webpage" />
</LinearLayout>
```

b. Change class **MainActivity** to the following.

```
package iss.nus.edu.sg.webviews.workshop

import android.content.Intent
import android.net.Uri
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import iss.nus.edu.sg.webviews.workshop.databinding.ActivityMainBinding

class MainActivity : AppCompatActivity() {
    private lateinit var binding: ActivityMainBinding

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        binding = ActivityMainBinding.inflate(layoutInflater)
        setContentView(binding.root)

        initButtons()
    }

    private fun initButtons() {
        binding.launchButton.setOnClickListener {
            val externalUrl = "https://nus.edu.sg"
            launchExternalPage(externalUrl)
        }
    }

    private fun launchExternalPage(externalUrl: String) {
        // To be implemented
        val intent = Intent(Intent.ACTION_VIEW, Uri.parse(externalUrl))
        startActivity(intent)
    }
}
```

4. Run the application. Make sure that:

- When the app launched, the **Main Activity** screen appears.
- When the “Launch External Webpage” button is clicked, the NUS home page is displayed.

Implement Features

Currently, we are using Implicit Intents to implement launching external webpages, which has some disadvantages. Use WebViews to implement the features described in the requirements.